

Kisan Setu — Change Request Document (CRD)

Manager Review Feedback & Proposed Modifications

Document Version: 1.1
Date: 27 August 2026
Status: Approved for Implementation
Source: Manager Review of Prototype v1.0

Executive Summary

Following the review of the Kisan Setu prototype application and PRD, the manager has approved 11 modifications spanning UI/UX theming, onboarding personalization, government scheme integration, marketplace intelligence, advisory AI, equipment booking, and land records search. This document formalizes each change with implementation context and technical notes.

Change Index

#	Change Title	Category
1	Font Update	Design System
2	Splash Screen —Group Logo Sequence	Branding / Onboarding
3	Light Color Theme Migration	Design System
4	Region-Wise Language Selection	Onboarding / Localization
5	Region-Wise Crop Suggestions	Onboarding / Advisory
6	Government Scheme Deep-Link Integration	Schemes / External Integration
7	Live Vyapari Rate Display on Dashboard	Marketplace / Dashboard
8	Chatbot Integration	Advisory / AI
9	Market Saturation Advisory (Demand Forecasting)	AI Advisory / Marketplace
10	Time-Slot-Based Yantra Booking	FPO / Equipment
11	7/12 Utara Search by Gat Number / Village	Land Records / Legal

Detailed Change Specifications

Change 1: Font Update

Category: Design System — Typography

Current State The design document specifies **Noto Sans** and **Noto Sans Devanagari** as the primary typeface across all text roles (Display, Heading, Body, Caption, Voice Response).

Proposed Change Replace the current font with a more culturally resonant, highly legible typeface optimized for rural Indian users and outdoor readability.

Recommended Options

Font	Rationale	Best For
Mukta (Google Fonts)	Open-source, designed for Indian languages, excellent Devanagari support, highly legible at small sizes	Primary replacement
Baloo 2	Friendly, rounded, approachable personality; great for headings and CTAs	Display & Headings
Hind	Clean, professional, optimized for screen readability in Indic scripts	Body text

Implementation Notes

- Ensure the selected font supports all 15+ target languages (Hindi, Marathi, Gujarati, Punjabi, Tamil, Telugu, Kannada, Bengali, Odia, Assamese, Malayalam).
- Maintain the existing typography scale (28sp Display down to 12sp Caption) — only change the font family.
- Test bold weights at 28sp for P&L hero numbers; ensure numerals (0–9) are highly distinguishable for semi-literate users.
- Verify rendering on entry-level Android devices (Android 6+, 2GB RAM) to avoid font-loading lag.

Impact

- Improved cultural connection and readability.
- No structural UI changes required.

Change 2: Splash Screen — Group of Companies Logo Sequence

Category: Branding / Onboarding

Current State The onboarding flow begins directly with the **Kisan Setu logo + tagline** splash screen.

Proposed Change Introduce a **two-step splash sequence**:

1. **Step 1:** Display the **Group of Companies logo** first (parent organization / consortium logo).
2. **Step 2:** Transition to the **Kisan Setu app logo** with tagline.

Implementation Notes

- **Sequence Timing:**
 - Group Logo: 2 seconds (static display with subtle fade-in).
 - Transition: Cross-fade or slide animation (300ms).
 - Kisan Setu Logo: 2 seconds, then auto-navigate to Language Selection.
- **Total splash duration:** ~4.5 seconds (within acceptable range for rural users with slower devices).
- **Offline Handling:** Both logos must be bundled in local assets (no network dependency).
- **Accessibility:** Respect `prefers-reduced-motion` — use instant cut instead of animation for users with motion sensitivity.

Visual Specification



Impact

- Reinforces brand trust by showing institutional backing.
- Aligns with corporate branding requirements.

Change 3: Light Color Theme Migration

Category: Design System — Color Palette

Current State The design uses a **mixed theme** with Primary Green (#2E7D32) headers, Earth Brown (#5D4037) grounding elements, and Charcoal (#212121) text on Cloud White (#FAFAFA) backgrounds. Some cards use darker tones.

Proposed Change Migrate the entire app to a **unified Light Color Theme** with softer, more approachable tones suitable for bright sunlight visibility and reduced eye strain.

Revised Color Palette

Token	Current Hex	New Light Theme Hex	Usage
Background	#FAFAFA	#F5F7FA	Screen backgrounds (soft blue-grey tint)
Card Background	#FFFFFF	#FFFFFF	Cards (pure white with subtle shadow)
Primary	#2E7D32	#43A047	Lighter, fresher green for actions

Table 3 – continued

Token	Current Hex	New Light Theme Hex	Usage
Primary Dark	#1B5E20	#2E7D32	Headers, emphasis (old primary becomes new dark)
Primary Light	#66BB6A	#81C784	Progress bars, highlights
Accent	—	#E8F5E9	Very light green for backgrounds, badges
Text Primary	#212121	#263238	Softer dark blue-grey instead of pure black
Text Secondary	#757575	#90A4AE	Muted blue-grey for placeholders
Surface	—	#FFFFFF	Elevated surfaces (cards, modals)
Divider	#5D4037	#ECEFF1	Light grey dividers instead of brown

Implementation Notes

- Ensure **WCAG 2.1 AA contrast ratios** are maintained with new colors:
 - New Primary (#43A047) on White: ~4.8:1 
 - Text Primary (#263238) on Background (#F5F7FA): ~12:1 
- Update all component tokens (buttons, cards, inputs, modals).
- Test on entry-level devices in **bright sunlight** — the light theme should reduce glare compared to dark elements.
- Update illustration style to match lighter, airier palette.

Impact

- Modern, clean aesthetic.
- Better outdoor readability.
- Requires full UI component audit and update.

Change 4: Region-Wise Language Selection

Category: Onboarding / Localization

Current State Language selection is a simple list/grid of languages with flag/audio preview icons. No regional grouping.

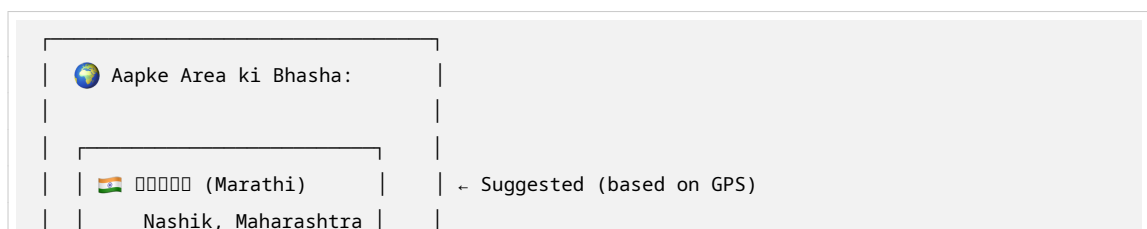
Proposed Change Organize language selection **by geographical region** rather than alphabetical order. Auto-detect user's state/region via GPS or manual district selection, then surface relevant languages first.

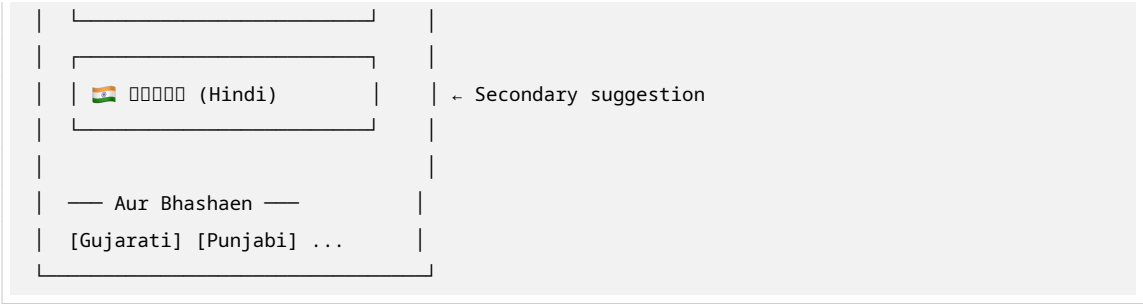
Regional Language Mapping

Region	States/UTs	Suggested Languages (Priority Order)
North India	Punjab, Haryana, Himachal, Delhi, Chandigarh	Punjabi, Hindi, Haryanvi
Central India	Uttar Pradesh, Madhya Pradesh, Chhattisgarh	Hindi, Bhojpuri, Bundelkhandi
West India	Maharashtra, Gujarat, Goa, Rajasthan	Marathi, Gujarati, Rajasthani, Hindi
East India	West Bengal, Bihar, Jharkhand, Odisha	Bengali, Odia, Hindi, Maithili
Northeast India	Assam, Meghalaya, Manipur, etc.	Assamese, Bengali, Hindi, English
South India	Tamil Nadu, Kerala, Karnataka, Andhra, Telangana	Tamil, Malayalam, Kannada, Telugu

Implementation Notes

- **Auto-Detection Flow:**
 1. Request location permission.
 2. Reverse-geocode to state/district.
 3. Pre-select the top 2 languages for that region at the top of the list.
 4. Show “More Languages” section below with all 15+ options.
- **Manual Override:** Always allow user to select any language, regardless of region.
- **UI Pattern:**





- **Audio Preview:** Each language card has a 🗣️ icon to hear “Namaste, Kisan Setu mein aapka swagat hai” in that language.

Impact

- Faster onboarding for farmers (one-tap language selection).
- Reduces cognitive load by prioritizing familiar options.

Change 5: Region-Wise Crop Suggestions

Category: Onboarding / Farm Profile / Advisory

Current State Crop selection during farm profile setup shows a generic list of crops. The PRD mentions regional prioritization but does not specify the mechanism.

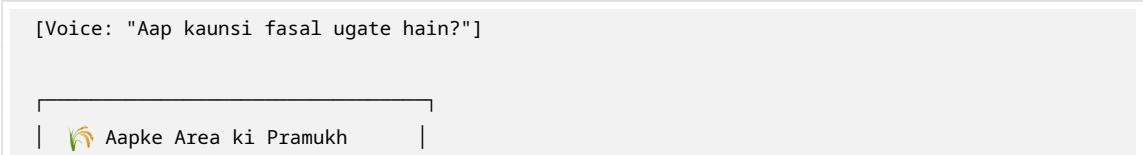
Proposed Change During **Farm Profile Setup** (voice-guided), suggest crops **based on the farmer’s selected district/region**. Use agro-climatic zone data + district-level cropping patterns.

Regional Crop Mapping (Examples)

State	District	Suggested Crops (Priority)
Maharashtra	Nashik	Tomato, Onion, Grape, Wheat
Maharashtra	Nagpur	Orange, Cotton, Soybean
Gujarat	Anand	Cotton, Tobacco, Wheat, Bajra
Gujarat	Banaskantha	Potato, Castor, Wheat
Punjab	Ludhiana	Wheat, Rice, Potato
Rajasthan	Sikar	Bajra, Mustard, Guar
UP	Lucknow	Wheat, Rice, Sugarcane, Potato
Tamil Nadu	Coimbatore	Cotton, Tomato, Coconut

Implementation Notes

- **Data Source:** Integrate with ICRISAT / State Agriculture Department district-level crop data.
- **UI Pattern:**



Fasalen:

 Tomato

 Onion

 Grape

 Wheat

Aur Fasalen

[Search or scroll all crops]

- **Smart Default:** Pre-select the top 2–3 crops based on district; allow multi-select.
- **Seasonal Awareness:** If onboarding happens in June (Kharif season), prioritize Kharif crops (cotton, soybean, rice) over Rabi crops (wheat, mustard).

Impact

- Reduces setup time.
- Increases relevance of early advisory and P&L calculations.
- Improves data accuracy for community verification features.

Change 6: Government Scheme — Direct Login Page Navigation

Category: Schemes / External Integration

Current State The PRD describes an in-app scheme application flow with auto-filled forms, status tracking, and document vault integration. The manager wants a **hybrid approach** for scheme access.

Proposed Change When a farmer **selects any government scheme** from the eligibility checker or scheme list, provide two options:

1. **“Apply via Kisan Setu”** (existing in-app flow with pre-filled forms).
2. **“Apply on Official Portal”** — redirects the user to the **official login page** of that specific government scheme.

Scheme-to-Portal Mapping

Scheme	Official Portal	Login Page URL
PM-KISAN	pmkisan.gov.in	pmkisan.gov.in/BeneficiaryStatus.aspx
PMFBY	pmfby.gov.in	pmfby.gov.in/ (farmer login)
Soil Health Card	soilhealth.dac.gov.in	soilhealth.dac.gov.in/
Kisan Credit Card (KCC)	Various bank portals	Bank-specific net banking / KCC portal
eNAM	enam.gov.in	enam.gov.in/web/
State-specific schemes	State agriculture portals	Variable

Implementation Notes

- **In-App Browser:** Use Chrome Custom Tabs (Android) or WebView with navigation controls.
- **Auto-Fill Injection:** Where technically possible, inject farmer's Aadhaar/land details into the official portal form fields using JavaScript bridge (requires portal cooperation).
- **Document Vault Integration:** Allow user to attach documents from the Kisan Setu vault directly to the official portal upload fields.
- **Deep-Link Handling:** If the official app is installed (e.g., PM-KISAN app), deep-link into it instead of browser.
- **Fallback:** If the portal is down, show a message: *“Sarkari website thodi der mein uplabdh hogi. Kripya baad mein koshish karein.”* and offer the in-app application as fallback.

Security & Compliance

- Ensure Aadhaar data is never stored locally in WebView cache.
- Clear session cookies after the user exits the external portal.
- Display a **security banner**: *“Aap sarkari portal par ja rahe hain. Kisan Setu aapka data surakshit rakhta hai.”*

Impact

- Builds trust by showing official government branding.
- Reduces Kisan Setu's liability for scheme application accuracy.
- Maintains user journey continuity.

Change 7: Live Vyapari Rate Display on Dashboard

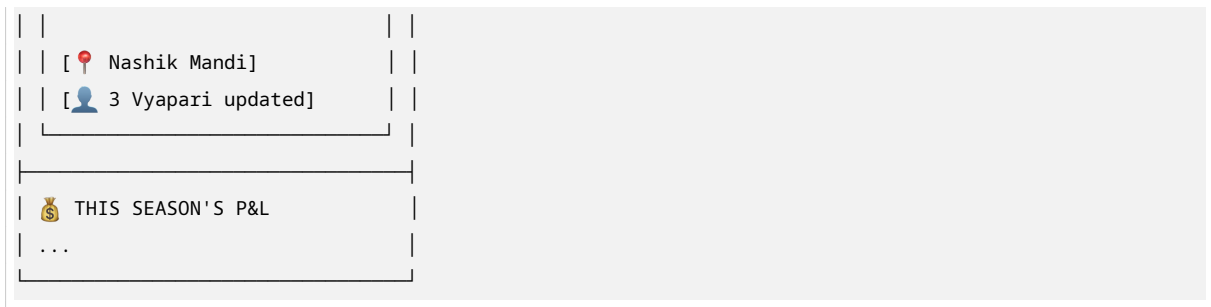
Category: Marketplace / Dashboard

Current State The dashboard shows P&L summary, weather, and quick actions. Mandi prices are accessed via a separate module or voice query.

Proposed Change Display **live crop rates decided by local vyaparis (traders)** directly on the **Home Dashboard** as a prominent card. This acts as a “Market Pulse” widget.

Dashboard Integration

[Greeting + Weather]		
[Tap to Speak]		
⚡ TODAY'S ACTION		
...		
💰 AAJ KE BHAV (Live Rates)		
🍅 Tomato:	₹24/kg	↑ ₹2
🧅 Onion:	₹18/kg	↓ ₹1
🌾 Wheat:	₹2,100/qt1	—



Implementation Notes

- **Data Source:**
 - Vyapari app/partner portal where traders input daily rates.
 - Integration with existing mandi price APIs (Agmarknet, eNAM) as fallback.
- **Update Frequency:** Every 2 hours during market hours (6 AM – 8 PM).
- **Visual Indicators:**
 - ↑ **Green:** Price increased vs. yesterday.
 - ↓ **Red:** Price decreased vs. yesterday.
 - — **Grey:** No change.
- **Crop Relevance:** Only show rates for crops the farmer has listed in their profile (configurable in settings).
- **Offline State:** Show last cached rates with timestamp: *“Kal ke bhav (internet nahi hai)”*.
- **Tap Action:** Tapping a crop rate opens the full Mandi Prices screen with trends, nearby mandis, and buyer offers.

Impact

- Increases dashboard utility and DAU (farmers check prices daily).
- Bridges the gap between trader pricing and farmer awareness.
- Supports the “Action Over Information” design principle.

Change 8: Chatbot Integration

Category: Advisory / AI

Current State The app has a **Voice Chat** interface (WhatsApp-style) for advisory queries. There is no persistent text-based chatbot widget.

Proposed Change Add a **dedicated Chatbot interface** accessible from:

1. **Floating Action Button (FAB)** on every screen (replaces or supplements the Voice FAB).
2. **Advisory Tab** as a primary entry point.

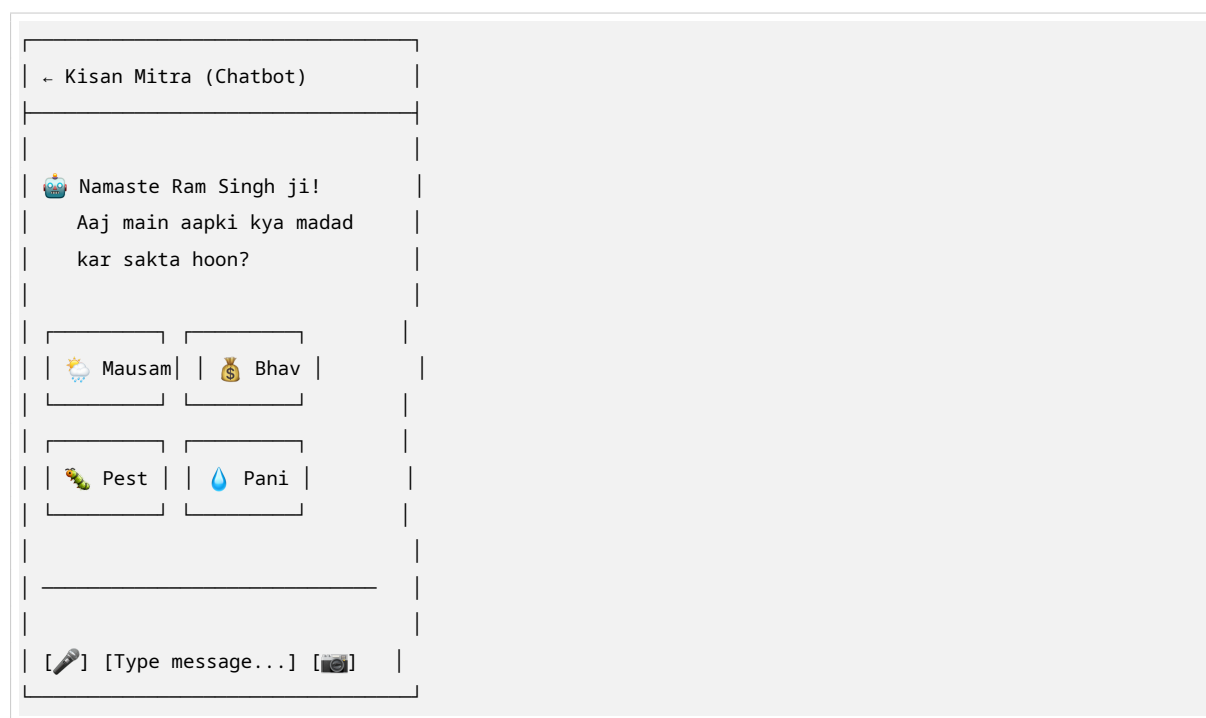
Chatbot Capabilities

Capability	Description	Example
Text + Voice Input	Farmers can type or speak queries	“Gehu mein peela dhabba”

Table 7 – continued

Capability	Description	Example
Quick Reply Buttons	Pre-set buttons for common questions	[Mausam] [Mandi Bhav] [Pest] [Pani]
Rich Cards	Visual responses with images, charts	Crop calendar card, P&L chart card
Multi-turn Conversations	Context-aware follow-up questions	Bot: “Kis fasal ki jankari chahiye?” → User: “Tamatar” → Bot: “Tamatar ka kya janna hai?”
Language Switching	Auto-detect or manual switch within chat	User switches from Hindi to Marathi mid-chat
Expert Handoff	Escalate to human expert with context	“Expert se baat karein” → passes chat history

UI Specification



Implementation Notes

- **Backend:** Use OpenRouter API (as per PRD tech stack) with fine-tuned agricultural prompts.
- **Context Memory:** Maintain session context for 24 hours (crop type, location, season).
- **Offline Mode:** Queue messages when offline; bot responds when connected.
- **Voice-to-Text:** Integrate Sarvam AI API for speech recognition in 15+ dialects.
- **Persona:** Friendly, respectful, knowledgeable — name it “**Kisan Mitra**”.
- **Accessibility:** Large text (20sp for bot responses), high contrast bubbles, audio playback for every response.

Impact

- Reduces dependency on voice-only navigation.
- Provides structured, visual advisory.
- Increases engagement through conversational UI.

Change 9: Market Saturation Advisory (Demand Forecasting)

Category: AI Advisory / Marketplace Intelligence

Current State The AI advisory engine provides pest diagnosis, weather alerts, and fertilizer recommendations. There is no market demand forecasting for crop selection.

Proposed Change When a farmer expresses intent to sow a specific crop (via voice or chatbot), the AI should analyze **local supply saturation** and provide a **demand-risk warning**.

User Flow Example **Farmer:** *“Is saal tamatar ugane ka soch raha hoon.”*

Bot Response:

⚠ MARKET JAANKARI

Aapke 5km ke aas-paas:

- 12 kisanon ne tamatar lagaya hai
- Pichle hafte mandi mein 200% zyada tamatar aaya

Anumanit Bhav (3 mahine baad):

₹12-15/kg (Gir sakta hai)

💡 Sujhav:

- Capsicum ki demand zyada hai
- Anumanit bhav: ₹25-30/kg

[Capsicum ke baare mein janein]

[Phir bhi Tamatar lagayein]

Data Inputs for Advisory

1. **Community Crop Data:** Aggregated (anonymized) sowing data from Kisan Setu users in 5km–20km radius.
2. **Mandi Arrival Data:** eNAM / Agmarknet arrival volumes for the crop.
3. **Historical Price Trends:** 3-year price data for the crop in the district.
4. **Export Demand:** Processor/exporter demand signals from the Direct-to-Buyer module.
5. **Weather Impact:** Climate suitability score for the crop this season.

Implementation Notes

- **Privacy:** Aggregate data only — never show “Farmer X is growing tomato.” Show counts (“12 farmers”).
- **Opt-in:** Farmers must opt-in to share anonymized crop data to receive this advisory.
- **Visual Design:** Use a **risk meter** (green/yellow/red) instead of complex charts.
 -  **Low Risk:** High demand, low supply.
 -  **Medium Risk:** Balanced market.
 -  **High Risk:** Oversupply predicted.
- **Alternative Suggestions:** Bot suggests 2–3 alternative crops with better demand prospects, linked to the region-wise crop database (Change 5).

Impact

- Helps farmers avoid price crashes due to oversupply.
- Increases income stability.
- Differentiates Kisan Setu from information-only apps.

Change 10: Time-Slot-Based Yantra (Equipment) Booking

Category: FPO / Equipment Sharing

Current State The PRD mentions a “Shared Equipment Calendar” (Module H3) but does not specify time-slot granularity.

Proposed Change Upgrade the **Yantra Booking** feature to support **time-slot-based reservations** instead of just full-day bookings. This maximizes equipment utilization and allows multiple farmers to share a machine in a single day.

Booking Interface

← Tractor Booking

 Date: 28 Aug 2026

 Time Slots:

 6:00 AM – 10:00 AM

 Available

[Book Slot]

 10:00 AM – 2:00 PM

 Booked (Ramesh Patel)

 2:00 PM – 6:00 PM

 Available

[Book Slot]

 6:00 PM – 10:00 PM

Pending Approval

View on Map

Call Owner

Time Slot Configuration (Default)

Slot	Duration	Typical Usage
Morning	6 AM –10 AM (4h)	Ploughing, tilling
Afternoon	10 AM –2 PM (4h)	Sowing, spraying
Evening	2 PM –6 PM (4h)	Harvesting, transport
Night	6 PM –10 PM (4h)	Irrigation, light work

Note: Configurable by equipment owner.

Implementation Notes

- **Calendar View:** Weekly view with color-coded availability (Green = Free, Red = Booked, Yellow = Pending).
- **Booking Rules:**
 - Max 2 slots per farmer per day (fair usage).
 - Cancellation allowed up to 2 hours before slot (SMS notification to owner).
 - Auto-confirmation for FPO-owned equipment; manual approval for private owners.
- **Reminders:** Push + SMS reminder 30 minutes before slot: “Aapka tractor slot 10 minute mein shuru hone wala hai.”
- **Pricing:** Per-slot pricing displayed upfront. Example: 800 for 4-hour tractor slot.
- **Waitlist:** If slot is full, farmer can join waitlist and get notified if cancelled.

Impact

- 3–4x better equipment utilization.
- Reduces farmer wait times during peak season.
- Increases FPO revenue from shared assets.

Change 11: 7/12 Utara Search by Gat Number or Village Name

Category: Land Records / Legal Toolkit

Current State The PRD mentions Bhulekh integration (Module K1) for viewing digital land records (7/12, Khatauni). The manager wants a **search-first approach** specific to Maharashtra’s 7/12 and 8A records.

Proposed Change In the **Land & Legal Toolkit** → **7/12 Utara** section, allow farmers to search their land record using:

1. **Gat Number** () — Survey/plot number.
2. **Village Name** () — Village name.

This replaces or supplements the current Aadhaar/land-owner-name-based search.

Search Interface (Maharashtra-Specific)

← 7/12 Utara (Maharashtra)

🔍 Apna 7/12 Kaise Dhundein?

 Gat Number se

 Village Name se

 Mera 8A bhi dekhein]

 Map par zameen dekhein]

Maharashtra 7/12 & 8A Reference

Document	What It Contains	Search Key
7/12 Utara	Ownership record, crop details, land area, liabilities	Gat Number + Village
8A Utara	Holder of rights, mutation status, tax details	Gat Number + Village
Malmatte Patrak	Property card for urban land	Survey Number

Implementation Notes

- **Data Source:** Integrate with **Maharashtra Bhulekh** (mahabhulekh.maharashtra.gov.in) or **Aaple Sarkar** APIs.
- **Search Flow:**
 1. User enters **Gat Number** (e.g., Gat No. 45) or **Village Name** (e.g., Niphad).
 2. System queries the Maharashtra land records database.
 3. If multiple records match, show a list: *“45 records mile. Aapka naam chunein.”*
 4. Display the 7/12 PDF with zoom, download, and share options.
- **Auto-Fill:** Once 7/12 is fetched, auto-extract:
 - Land area (in acres/guntha).

- Owner name.
- Crop history (last season).
- Soil type (if mentioned). → Pre-fill into Farm Profile and P&L Dashboard.
- **8A Integration:** Toggle switch to view 8A record for the same Gat Number.
- **Offline:** Download and store 7/12 PDF in Document Vault for offline access.
- **Other States:** Build the framework to support similar searches for other states (e.g., Gujarat's 7/12, Karnataka's RTC, UP's Khatauni) using the same UI pattern but different API endpoints.

Validation Rules

- Gat Number: Alphanumeric, 1–20 characters.
- Village Name: Minimum 3 characters, fuzzy search enabled (handles spelling variations).
- District auto-detected from profile; allow manual override.

Impact

- Faster land record access for Maharashtra farmers.
- Reduces dependency on patwaris/village offices.
- Enables accurate farm profile auto-population.

Implementation Roadmap

Phase	Changes	Timeline	Dependencies
Sprint 1	Change 1 (Font), Change 2 (Splash), Change 3 (Light Theme)	Week 1–2	Design assets ready
Sprint 2	Change 4 (Language), Change 5 (Crops)	Week 3–4	Regional data API
Sprint 3	Change 6 (Scheme Portal), Change 8 (Chatbot)	Week 5–7	Portal partnerships, OpenRouter setup
Sprint 4	Change 7 (Vyapari Rates), Change 9 (Saturation)	Week 6–8	Vyapari partner onboarding, demand data
Sprint 5	Change 10 (Yantra Slots), Change 11 (7/12 Search)	Week 8–10	Equipment owner app, Bhulekh API

Updated Design Deliverables Checklist

- **Revised Typography Scale** — New font family, all languages tested.
- **Light Theme Color Tokens** — Updated palette in Figma/Design System.
- **Splash Screen Flow** — Group logo + Kisan Setu logo sequence.
- **Region-Wise Language Selection Screen** — GPS-based suggestions.

- **Region-Wise Crop Selection Screen** — District-based smart defaults.
 - **Scheme Portal Integration Spec** — Deep-linking, WebView, security flow.
 - **Dashboard Live Rate Card** — Vyapari rate widget design.
 - **Chatbot UI Kit** — Message bubbles, quick replies, rich cards, voice input.
 - **Market Saturation Advisory Card** — Risk meter, alternative crop suggestions.
 - **Yantra Time-Slot Booking Calendar** — Daily/weekly slot views.
 - **7/12 Search Interface** — Gat Number / Village Name input, results view.
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Document End